

Brendan Birozy

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EDUCATION

California State Polytechnic University, Pomona

Expected May 2027

B.S. Electrical Engineering - In Progress, **GPA 3.65**; Dean's List

- Microcontrollers, Microelectronics, Signals and Systems, Control Systems, Electromagnetic Fields, C++, Digital Logic, Circuit Analysis

EXPERIENCE

Formula SAE Electric, Cal Poly Pomona

Aug 2025 - Present

Electrical Design Engineer, EV Powertrain

- Develop High Voltage (400V) and Low Voltage (12V, 5V, 3.3V) Battery Management System (BMS) using KiCAD, ensuring designs meet safety and performance rules of FSAE national competition.
- Develop shutdown, precharge/discharge, indicator light, sensor, and communications BMS circuitry.
- Implement CAN and UART communications utilizing Teensy4.1 and perform analysis to verify robust, cost-effective component selections.
- Integrate BMS with 96s4p (six 16s4p P50B cell modules in series) Tractive Battery and Cascadia PM100DX + Emrax 228MV powertrain as part of a 10-member powertrain sub-team and greater 30-member multidisciplinary engineering team.

System Lead, Business and Marketing

- Lead sponsorship strategy by researching, identifying, and securing partnerships with industry professionals and alumni to support team growth and operations.
- Organize and lead a team of 3 to draft and compete in business proposal and cost analysis events, striving to maximize team ROI across all subteams.
- Direct weekly student engagement initiatives to strengthen recruitment and retention while managing and revamping social media utilizing LinkedIn, Canva, Instagram, and Adobe products, increasing brand visibility and following by 20%.
- Build continuously update/maintain team website to promote brand image utilizing Framer web development suite.

Autonomous Vehicle Laboratory, Cal Poly Pomona

Aug 2025 - Present

Embedded Systems Team Member

- Design and prototype control systems for autonomous drive-by-wire systems (steering, throttle, and braking) and migrate to permanent PCBs using KiCAD.
- Work with 20-person time to design a go-kart sized autonomous vehicle's drive-by-wire electrical system, integrating steering, throttle, and braking over a CAN bus network with Teensy4.1s.
- Research custom 4WD hub motor powertrain strategies for an autonomous EV project with off-road applications.

SKILLS

- Hardware: Circuit Analysis, Microcontrollers, Microelectronics, Digital Logic, Soldering, Lab Equipment
- CAN, SPI, C/C++, Orcad, KiCAD, PSpice, LTSpice, MATLAB, Simulink, NI LabVIEW, Verilog, Microsoft Office, Google Suite
- Languages: English [Native/Fluent], Japanese [Novice/Student]